

Frozen grouped factor model (direct HDBSCAN IS 2012-2022)

Directed factor-mimicking portfolio catalogue

Abstract

This catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by reported raw Sharpe, highest first. Direction is frozen upstream; this report performs no return-based sign selection.

Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes the reported Sharpe order and feeds all reported analyses. Displayed signal signs define the pre-direction composite; the reported Direction then orients the whole FMP. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only. Return Uncertainty uses paired circular stationary-bootstrap draws of the common monthly panel. It measures sampling and path uncertainty, not out-of-sample generalisation.

Construction Configuration

Setting	Effective value
Factors	7
Universe	shared security screen; refreshed monthly
Formation filters	price $\geq$ None; 21d dollar volume $\geq$ none
Market-cap screen	top 90.0% of names monthly
Signal preparation	within-month winsorize, normalize, then demean
Composite	mean available normalized components; factor-specific coverage floor
Construction	one blanket profile
Rebalance / return	month-end / next calendar month simple return
Costs	effective configured costs; default 0 bps
Minimum sample	20 stocks/month; 24 common months; 90% factor coverage
Comparison window	2012-02-01 to 2022-12-01; factor- and control-complete
Winsor settings	1%-99%
Controls	MKT, SMB, VAL, MOM, BAB, QMJ
Return uncertainty	paired stationary bootstrap; 2000 draws; expected block 12 months; seed 0

Returns PCA

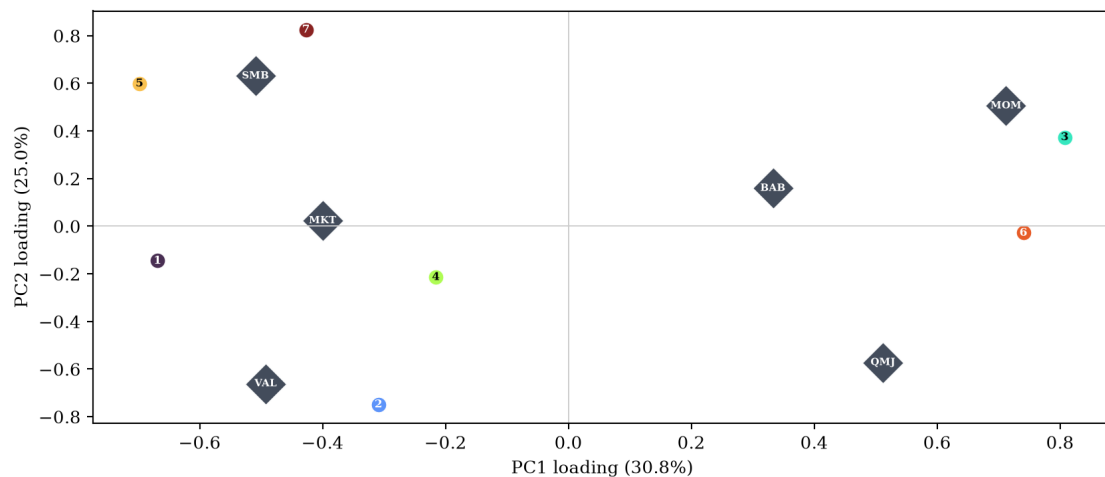


Figure: catalogue factors are numbered by reported-Sharpe index; attribution controls are named; $N=131$ common months.

Index

Idx	Factor	Dir.	N
1	Factor_5	+1	131
2	Factor_1	+1	131
3	Factor_7	+1	131
4	Factor_3	+1	131
5	Factor_2	+1	131
6	Factor_4	+1	131
7	Factor_6	+1	131

Returns

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Factor_5	8.01%	6.37%	1.26	4.42	-4.65%	0.294	9.92%	6.37%
Factor_1	4.75%	8.18%	0.58	1.68	-20.62%	0.703	2.09%	8.18%
Factor_7	4.15%	8.77%	0.47	1.78	-14.95%	0.677	-0.83%	8.77%
Factor_3	2.69%	6.28%	0.43	1.46	-10.48%	0.144	4.09%	6.28%
Factor_2	4.17%	11.06%	0.38	1.06	-21.53%	0.735	7.37%	11.06%
Factor_4	2.19%	5.86%	0.37	1.09	-18.01%	0.300	-1.39%	5.86%
Factor_6	1.97%	8.97%	0.22	0.63	-30.70%	0.627	4.84%	8.97%

Return Uncertainty

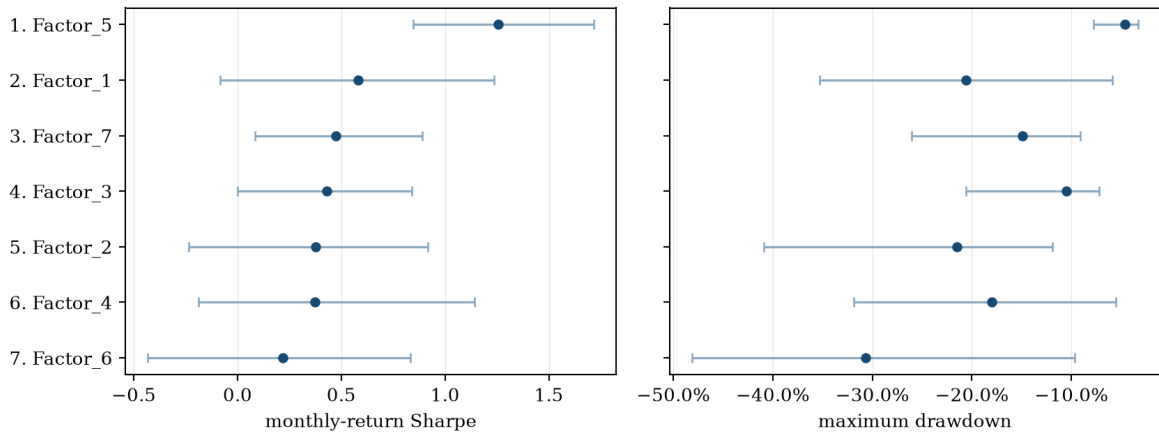


Figure: point estimates and paired stationary-bootstrap 95% intervals.

Idx	Factor	Ann. ret. [95%]	Sharpe [95%]	Pr(S<=0)	Skew [95%]	Max DD [95%]	ES 5% [95%]
1	Factor_5	8.0% [4.8%, 11.8%]	1.26 [0.85, 1.72]	0.0%	0.94 [0.21, 1.31]	-4.7% [-7.8%, -3.3%]	-2.2% [-2.3%, -1.9%]
2	Factor_1	4.8% [-0.7%, 10.3%]	0.58 [-0.08, 1.24]	4.5%	0.36 [-0.74, 1.06]	-20.6% [-35.3%, -5.9%]	-4.6% [-5.9%, -2.8%]
3	Factor_7	4.1% [0.7%, 7.4%]	0.47 [0.08, 0.89]	0.9%	-0.48 [-0.79, -0.09]	-15.0% [-26.0%, -9.1%]	-5.8% [-6.8%, -4.1%]
4	Factor_3	2.7% [-0.0%, 5.5%]	0.43 [-0.00, 0.84]	2.5%	1.26 [-0.35, 2.49]	-10.5% [-20.6%, -11.9%]	-3.3% [-4.1%, -2.3%]
5	Factor_2	4.2% [-2.1%, 12.3%]	0.38 [-0.23, 0.92]	11.5%	1.21 [-0.08, 1.61]	-21.5% [-40.9%, -11.9%]	-5.0% [-5.9%, -3.9%]
6	Factor_4	2.2% [-1.3%, 5.2%]	0.37 [-0.19, 1.14]	11.4%	-1.38 [-1.84, -0.29]	-18.0% [-31.9%, -5.5%]	-4.2% [-6.1%, -2.4%]
7	Factor_6	2.0% [-3.7%, 8.4%]	0.22 [-0.43, 0.83]	23.7%	0.73 [-0.80, 1.55]	-30.7% [-48.1%, -9.7%]	-5.5% [-6.2%, -3.0%]

Risk Loadings

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Factor_5	0.13 (3.13)	0.06 (1.02)	0.02 (0.36)	-0.19 (-3.39)	-0.07 (-0.94)	-0.05 (-0.85)	8.16% (5.22)	0.343
Factor_1	0.12 (4.22)	-0.06 (-1.05)	0.46 (12.85)	-0.09 (-2.05)	0.16 (3.63)	0.25 (4.32)	1.88% (1.17)	0.704
Factor_7	0.04 (1.15)	-0.12 (-2.06)	-0.17 (-2.68)	0.48 (7.72)	0.13 (2.45)	0.03 (0.43)	-0.58% (-0.44)	0.678
Factor_3	0.09 (1.36)	-0.11 (-1.91)	0.09 (1.96)	0.01 (0.08)	-0.19 (-3.26)	-0.01 (-0.14)	3.33% (1.65)	0.158
Factor_2	-0.02 (-0.44)	0.93 (11.40)	0.07 (1.32)	-0.06 (-1.05)	0.05 (0.51)	-0.32 (-3.57)	7.05% (4.01)	0.736
Factor_4	0.08 (1.86)	-0.10 (-1.47)	-0.07 (-1.29)	0.17 (3.13)	0.07 (0.83)	0.15 (1.57)	-1.36% (-0.81)	0.300
Factor_6	-0.06 (-1.31)	0.44 (5.86)	-0.13 (-2.21)	0.05 (0.92)	0.09 (1.09)	-0.51 (-5.32)	4.63% (2.36)	0.628

Return Contributions

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Factor_5	1.72%	-0.05%	-0.04%	-0.93%	-0.58%	-0.26%	8.16%	8.01%
Factor_1	1.49%	0.05%	-1.06%	-0.44%	1.40%	1.42%	1.88%	4.75%
Factor_7	0.49%	0.10%	0.38%	2.43%	1.15%	0.18%	-0.58%	4.15%
Factor_3	1.19%	0.09%	-0.20%	0.03%	-1.68%	-0.06%	3.33%	2.69%
Factor_2	-0.26%	-0.82%	-0.16%	-0.29%	0.45%	-1.81%	7.05%	4.17%
Factor_4	0.98%	0.09%	0.16%	0.88%	0.60%	0.86%	-1.36%	2.19%
Factor_6	-0.71%	-0.38%	0.30%	0.27%	0.76%	-2.90%	4.63%	1.97%

1. Factor_5

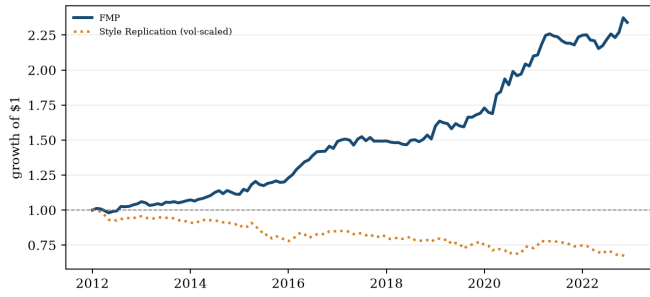


Figure: Common-window cumulative growth of \$1.

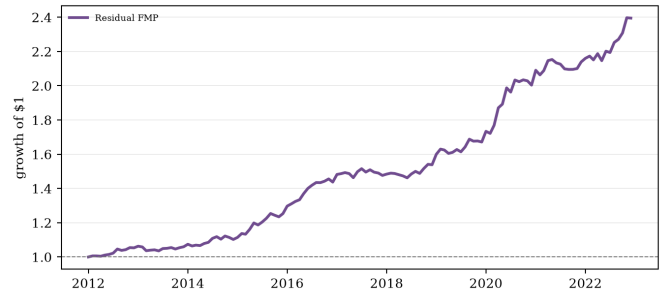


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Factor_5	8.01%	6.37%	1.26	4.42	-4.65%	0.294	9.92%	6.37%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Factor_5	0.13 (3.13)	0.06 (1.02)	0.02 (0.36)	-0.19 (-3.39)	-0.07 (-0.94)	-0.05 (-0.85)	8.16% (5.22)	0.343

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Factor_5	1.72%	-0.05%	-0.04%	-0.93%	-0.58%	-0.26%	8.16%	8.01%

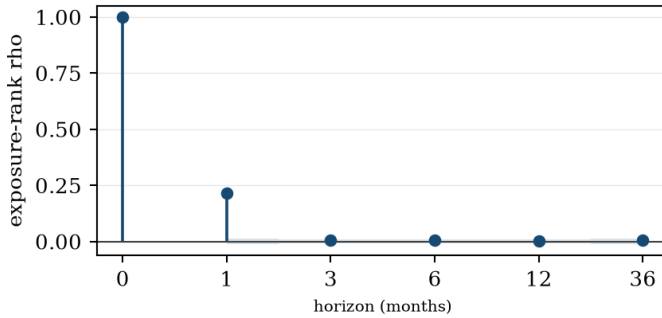


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

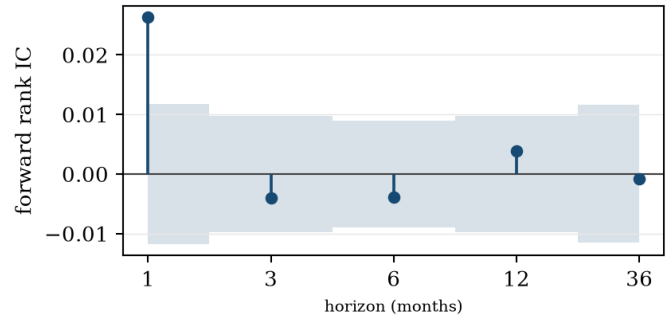


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	price_zscore_63d(-1), bb_pctb_20d(-1), ma_dist_63d(-1), momentum_0_21(-1), resid_ret_252v_1260c_SPY(-1), ret_volume_corr_63d(-1), ret_zscore_63d(-1), rsi_14d(-1), skew_63d(-1), stoch_k_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	6 of 10
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	1.26
Bootstrap Sharpe	1.26 [0.85, 1.72]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.94 [0.21, 1.31]; max DD=-4.7% [-7.8%, -3.3%]

2. Factor_1

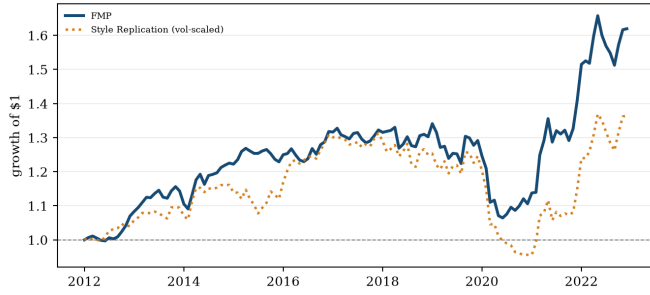


Figure: Common-window cumulative growth of \$1.

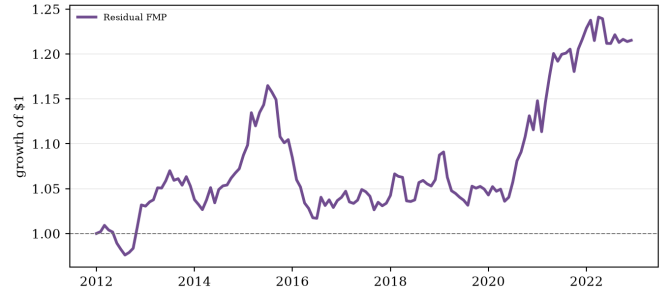


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Factor_1	4.75%	8.18%	0.58	1.68	-20.62%	0.703	2.09%	8.18%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Factor_1	0.12 (4.22)	-0.06 (-1.05)	0.46 (12.85)	-0.09 (-2.05)	0.16 (3.63)	0.25 (4.32)	1.88% (1.17)	0.704

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Factor_1	1.49%	0.05%	-1.06%	-0.44%	1.40%	1.42%	1.88%	4.75%

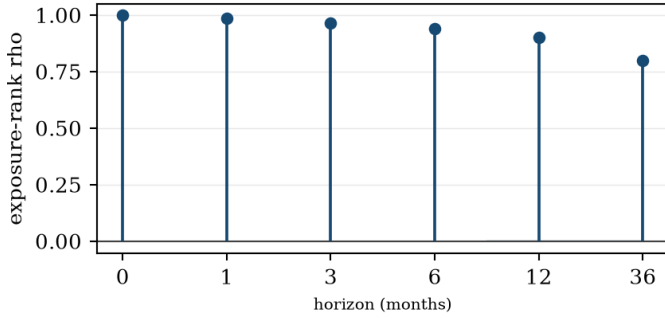


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

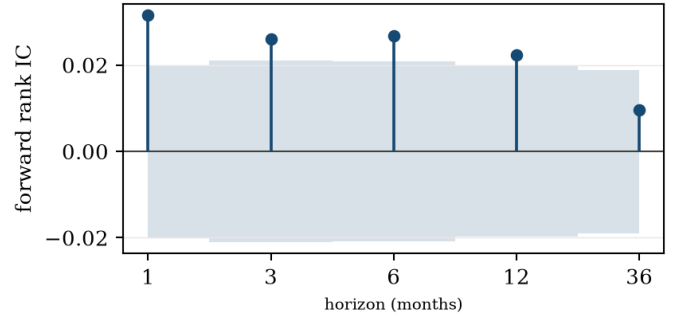


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	sales_to_price(+1), ps_ttm(-1), pb(-1), ncav_to_mc(-1), gross_profit_to_ev(+1), gpoa(+1), gp_to_equity(+1), gp_ev(+1), fcf_to_market_cap(+1), roic_nopat(-1), altman_z(-1), evs_ttm(-1), ev_to_assets(-1), debt_to_equity(+1), debt_to_assets(+1), current_ratio(-1), cash_ratio(-1), assets_to_mc(+1), asset_turnover(+1), fcf_margin(-1), fcf_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	13 of 21
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1237 / 1563 / 2078
Available-history Sharpe	0.58
Bootstrap Sharpe	0.58 [-0.08, 1.24]; Pr(S<=0)=4.5%
Bootstrap tails	skew=0.36 [-0.74, 1.06]; max DD=-20.6% [-35.3%, -5.9%]

3. Factor_7

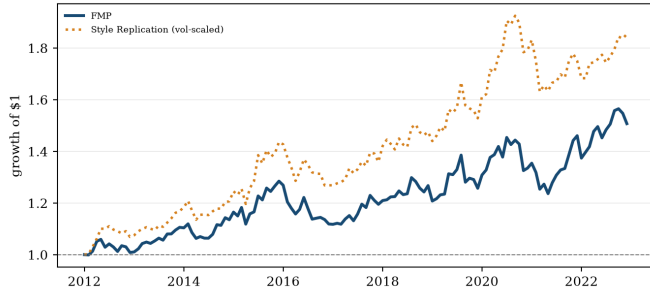


Figure: Common-window cumulative growth of \$1.

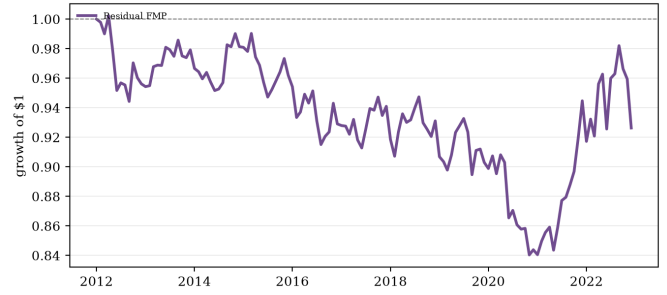


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Factor_7	4.15%	8.77%	0.47	1.78	-14.95%	0.677	-0.83%	8.77%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Factor_7	0.04 (1.15)	-0.12 (-2.06)	-0.17 (-2.68)	0.48 (7.72)	0.13 (2.45)	0.03 (0.43)	-0.58% (-0.44)	0.678

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Factor_7	0.49%	0.10%	0.38%	2.43%	1.15%	0.18%	-0.58%	4.15%

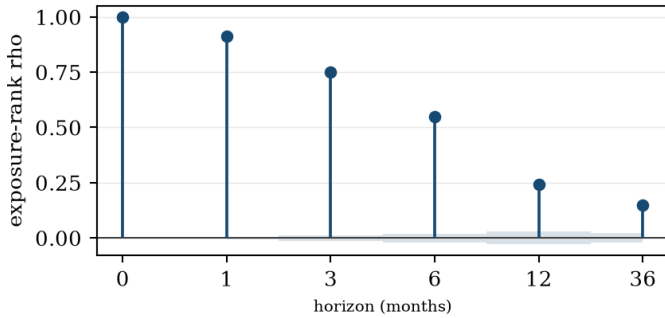


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

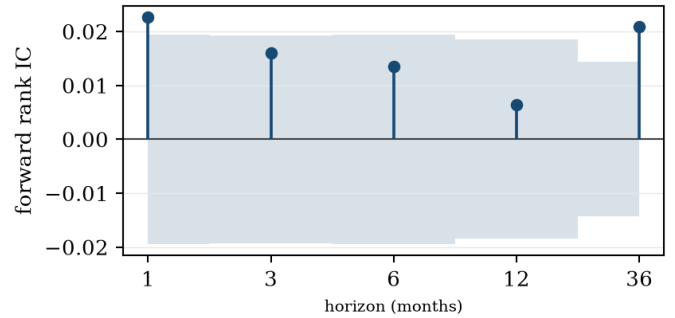


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	resid_momentum_21_252(+1), momentum_21_126(+1), momentum_21_252(+1), momentum_21_252_vol_63d(+1), momentum_21_756(+1), value_proxy_1260d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	4 of 6
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	0.47
Bootstrap Sharpe	0.47 [0.08, 0.89]; Pr(S<=0)=0.9%
Bootstrap tails	skew=-0.48 [-0.79, -0.09]; max DD=-15.0% [-26.0%, -9.1%]

4. Factor_3

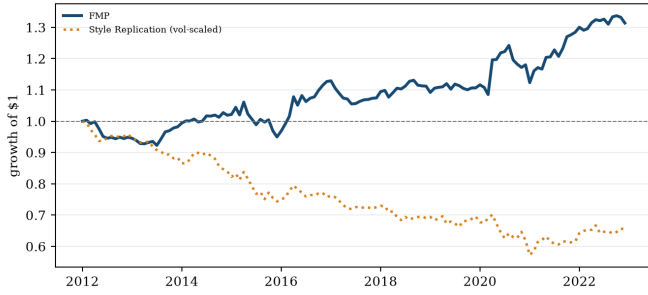


Figure: Common-window cumulative growth of \$1.

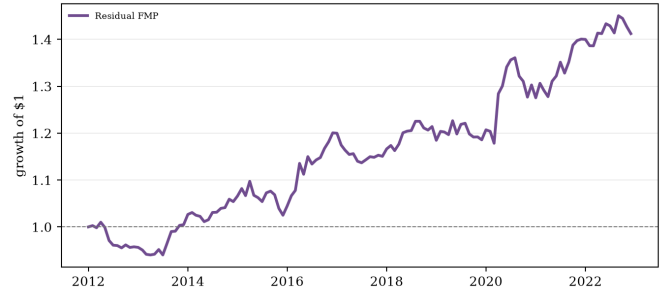


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Factor_3	2.69%	6.28%	0.43	1.46	-10.48%	0.144	4.09%	6.28%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Factor_3	0.09 (1.36)	-0.11 (-1.91)	0.09 (1.96)	0.01 (0.08)	-0.19 (-3.26)	-0.01 (-0.14)	3.33% (1.65)	0.158

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Factor_3	1.19%	0.09%	-0.20%	0.03%	-1.68%	-0.06%	3.33%	2.69%

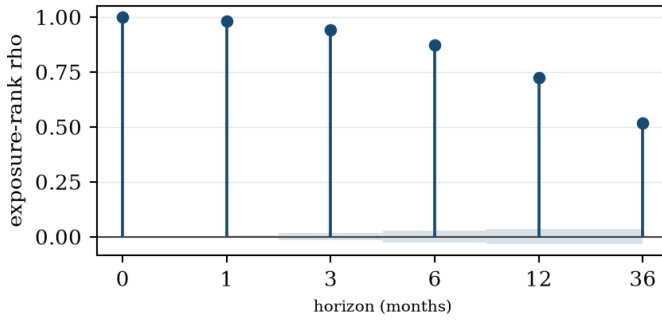


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

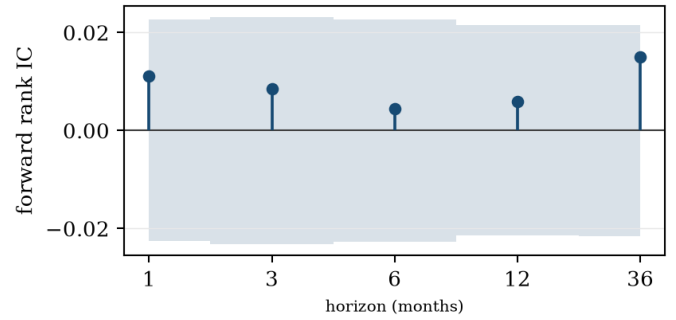


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	beta_252v_1260c_FRENCH_CRSP(-1), bab_252v_1260c_FRENCH_CRSP(+1), bab_252v_1260c_SPY(+1), bab_252v_1260c_VTI(+1), bab_shrunk_252v_1260c_FRENCH_CRSP(+1), bab_shrunk_252v_1260c_SPY(+1), bab_shrunk_252v_1260c_VTI(+1), beta_252v_1260c_SPY(-1), beta_252v_1260c_VTI(-1), beta_shrunk_252v_1260c_FRENCH_CRSP(-1), beta_shrunk_252v_1260c_SPY(-1), beta_shrunk_252v_1260c_VTI(-1), lottery_max_5_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta_neutral=True; group=sector; cost=0bp
Components required	8 of 13
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.43
Bootstrap Sharpe	0.43 [-0.00, 0.84]; Pr(S<=0)=2.5%
Bootstrap tails	skew=1.26 [-0.35, 2.49]; max DD=-10.5% [-20.6%, -7.2%]

5. Factor_2

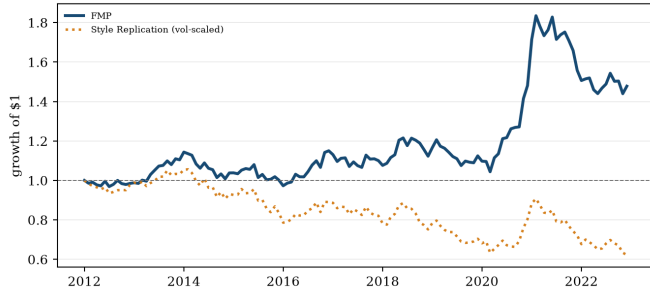


Figure: Common-window cumulative growth of \$1.

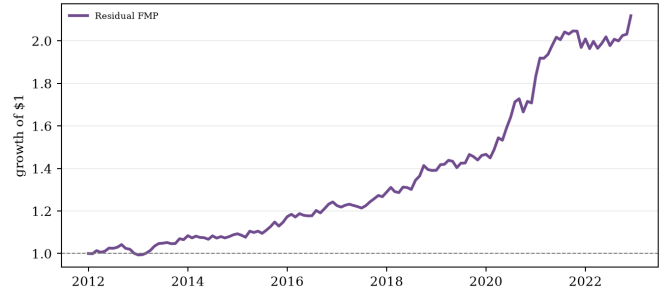


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Factor_2	4.17%	11.06%	0.38	1.06	-21.53%	0.735	7.37%	11.06%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Factor_2	-0.02 (-0.44)	0.93 (11.40)	0.07 (1.32)	-0.06 (-1.05)	0.05 (0.51)	-0.32 (-3.57)	7.05% (4.01)	0.736

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Factor_2	-0.26%	-0.82%	-0.16%	-0.29%	0.45%	-1.81%	7.05%	4.17%

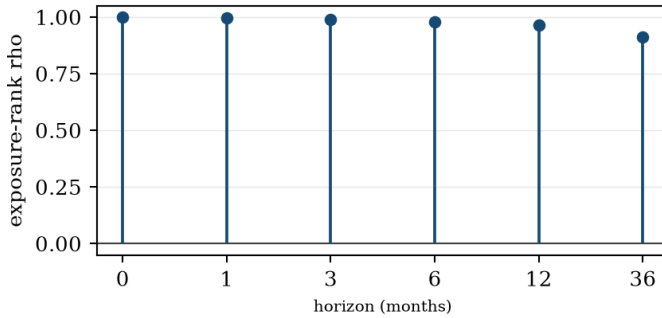


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

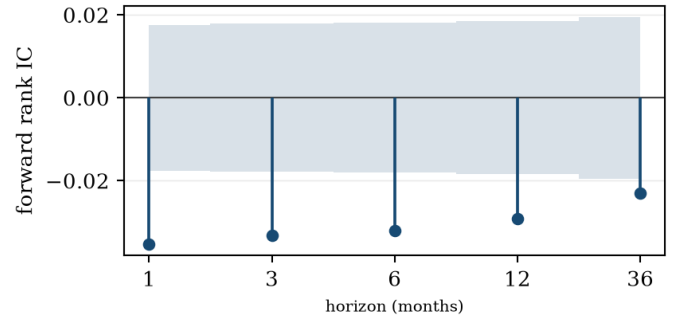


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	log_market_cap(-1), accruals(-1), accruals_percent(+1), accruals_total(-1), amihud_illiq_21d(+1), cash_earnings_ttm(-1), corr_1260d_FRENCH_CRSP(-1), corr_1260d_SPY(-1), corr_1260d_VTI(-1), dollar_volume_21d(-1), ev(-1), growth_acc_5y(-1), market_cap(-1), net_operating_assets_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	9 of 14
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1256 / 1569 / 2093
Available-history Sharpe	0.38
Bootstrap Sharpe	0.38 [-0.23, 0.92]; Pr(S<=0)=11.5%
Bootstrap tails	skew=1.21 [-0.08, 1.61]; max DD=-21.5% [-40.9%, -11.9%]

6. Factor_4

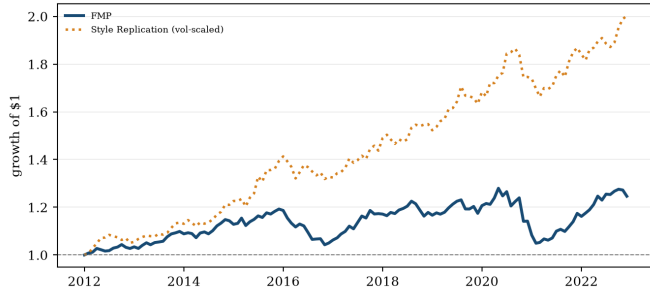


Figure: Common-window cumulative growth of \$1.

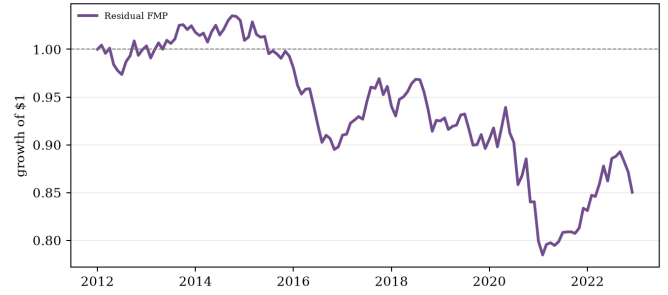


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Factor_4	2.19%	5.86%	0.37	1.09	-18.01%	0.300	-1.39%	5.86%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Factor_4	0.08 (1.86)	-0.10 (-1.47)	-0.07 (-1.29)	0.17 (3.13)	0.07 (0.83)	0.15 (1.57)	-1.36% (-0.81)	0.300

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Factor_4	0.98%	0.09%	0.16%	0.88%	0.60%	0.86%	-1.36%	2.19%

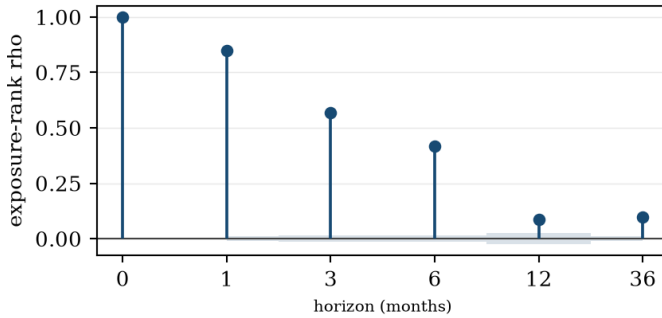


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

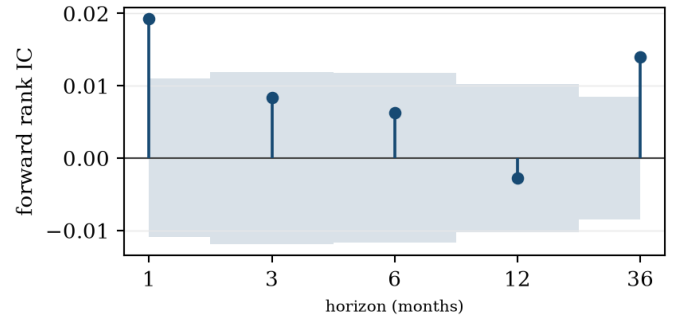


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	eps_growth_yoy(+1), accruals_noa_proxy(+1), asset_growth_yoy(+1), debt_issuance_yoy(-1), ebit_growth_yoy_ttm(+1), equity_growth_yoy(+1), net_income_growth_yoy(+1), net_income_growth_yoy_sign_flip(+1), revenue_growth_yoy(+1), surprise(+1), surprisePercentage(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	7 of 11
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1252 / 1564 / 2086
Available-history Sharpe	0.37
Bootstrap Sharpe	0.37 [-0.19, 1.14]; Pr(S<=0)=11.4%
Bootstrap tails	skew=-1.38 [-1.84, -0.29]; max DD=-18.0% [-31.9%, -5.5%]

7. Factor_6

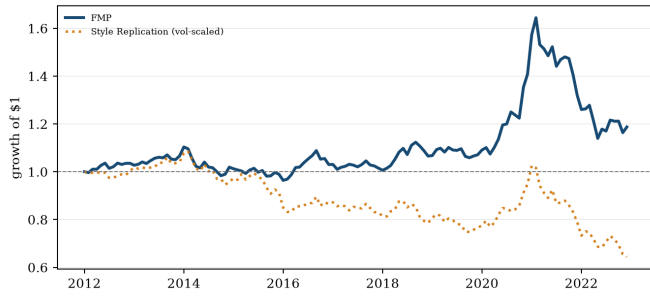


Figure: Common-window cumulative growth of \$1.

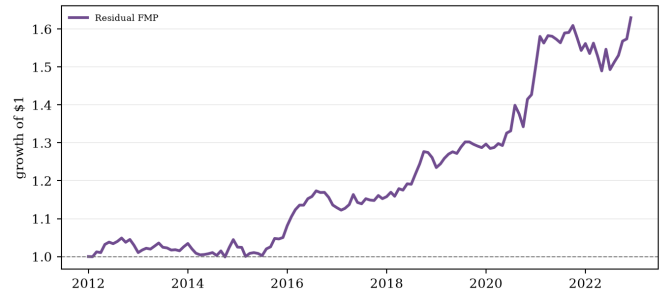


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
Factor_6	1.97%	8.97%	0.22	0.63	-30.70%	0.627	4.84%	8.97%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
Factor_6	-0.06 (-1.31)	0.44 (5.86)	-0.13 (-2.21)	0.05 (0.92)	0.09 (1.09)	-0.51 (-5.32)	4.63% (2.36)	0.628

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
Factor_6	-0.71%	-0.38%	0.30%	0.27%	0.76%	-2.90%	4.63%	1.97%

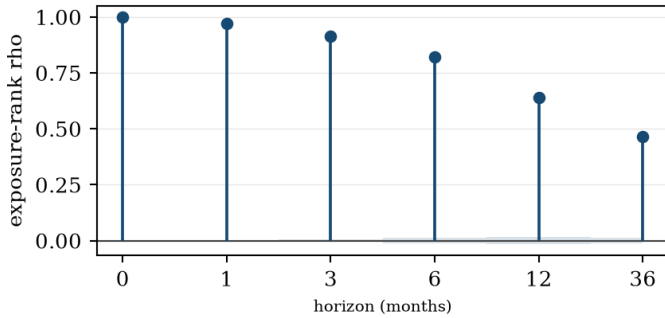


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

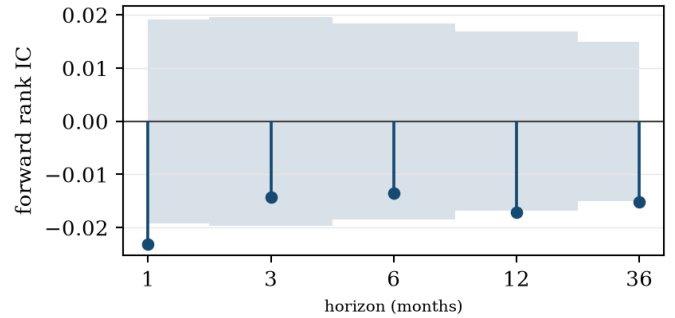


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	earnings_yield(-1), ebit_margin_ttm(-1), ebit_to_ev(-1), net_income_margin_ttm(-1), net_income_to_market_cap(-1), pe_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	4 of 6
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1213 / 1510 / 2029
Available-history Sharpe	0.22
Bootstrap Sharpe	0.22 [-0.43, 0.83]; Pr(S<=0)=23.7%
Bootstrap tails	skew=0.73 [-0.80, 1.55]; max DD=-30.7% [-48.1%, -9.7%]

Reproducibility

Run ID: 20260906T070257Z_frozen-grouped-factor-model-direct-hdbscan-is-2012-2022_d48ee472de.

Gold snapshot: version=20260829T051952Z.

Configuration hash: d48ee472de.

Symbols requested / loaded: 0 / 3009.

Seed: 0. Comparison months: 131, 2012-02-01 to 2022-12-01.

Command:

C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/automatic_factor_grouping/direct_handoff.py--source-rundata/outputs/automatic_factor_grouping/runs/20260906T065013Z_25145edccf.

Direction is frozen upstream; this report performs no return-based sign selection. The reported direction is applied consistently before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.